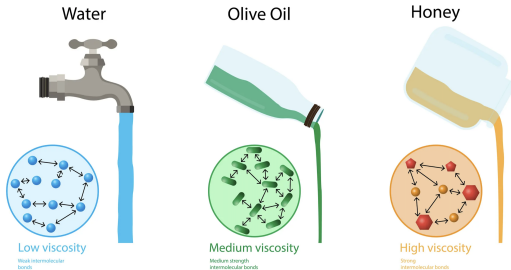


Nonlinear PDEs: Burgers' Equation

A slight modification of the heat equation yields a simplified fluid dynamics model, .

Definition

Burgers' equation is given by $u_t + uu_x = Du_{xx}$, where $D > 0$ introduces **viscosity**.





Burgers' Equation

The full Burgers' equation with **Dirichlet** boundary conditions is given by:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = D \frac{\partial^2 u}{\partial x^2}$$

$$u(x, 0) = f(x), \quad x \in [a, b] \quad \text{initial temperature profile}$$

$$u(a, t) = l(t), \quad t \in [t_0, T] \quad \text{temp on the left boundary}$$

$$u(b, t) = r(t), \quad t \in [t_0, T] \quad \text{temp on the right boundary}$$



Solving with Finite Difference Method

Approximate the left-hand side of Burgers' equation

$u_t + uu_x = Du_{xx}$ using a backward difference for u_t and a centered difference for u_x :

$$\left[\frac{u_{i,j} - u_{i,j-1}}{k} + \mathcal{O}(k) \right] + u_{ij} \left[\frac{u_{i+1,j} - u_{i-1,j}}{2h} + \mathcal{O}(h^2) \right].$$

The right-hand side is approximated as before:

$$D \left[\frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2} + \mathcal{O}(h^2) \right]$$

Due to the nonlinear term uu_x , we cannot write the system of equations as a linear system, e.g., $\mathbf{A}\mathbf{u}_j = \mathbf{b}_{j-1}$



Solving with Finite Difference Method

Letting $\sigma = Dk/h^2$, the nonlinear system of equations takes the form:

$$u_{i,j} + \frac{k}{2h} u_{i,j} [u_{i+1,j} - u_{i-1,j}] - \sigma [u_{i+1,j} - 2u_{i,j} + u_{i-1,j}] - u_{i,j-1} = 0.$$

Letting $\mathbf{u}_j = [u_{0,j}, u_{1,j}, \dots, u_{n,j}]^T$, note that $\mathbf{u}_j$ must solve the nonlinear equation

$$0 = f_i(\mathbf{z}) = z_i + \frac{k}{2h} z_i [z_{i+1} - z_{i-1}] - \sigma [z_{i+1} - 2z_i + z_{i-1}] - u_{i,j-1},$$

where $u_{i,j-1}$ is known from the previous time step.

(See notes.)



Solving with Finite Difference Method

We can use a root-finding algorithm to propagate the approximate solution $\mathbf{u}_{j-1}$ at time t_{j-1} to obtain $\mathbf{u}_j$ at time t_j . (Or equivalently, from time t_j to time t_{j+1} .)

Let $F(\mathbf{z}) = [f_0(\mathbf{z}), f_1(\mathbf{z}), \dots, f_{n-1}(\mathbf{z}), f_n(\mathbf{z})]^T$ contain the functions and note that the boundary conditions require different equations:

$$0 = f_0(\mathbf{z}) = z_0 - l(t_j)$$

$$0 = f_n(\mathbf{z}) = z_n - r(t_j)$$

(See notes.)



Recall Newton's Root-finding Method

It is typically preferable to find the root using Newton's method:

Definition

Newton's method for root finding uses the iterative process

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} - [J(\mathbf{x}^{(t)})]^{-1} F(\mathbf{x}^{(t)}), \quad (\text{vector } \mathbf{x})$$

where $J(\mathbf{x})$ is the **Jacobian matrix** for vector function $F(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_n(\mathbf{x})]$ in which $J_{ij}(\mathbf{x}) = \frac{\partial}{\partial x_j} f_i(\mathbf{x})$.

In practice, it's best to not invert J but instead solve the linear system:

$$J(\mathbf{x}^{(t)})\mathbf{x}^{(t+1)} = J(\mathbf{x}^{(t)})\mathbf{x}^{(t)} - F(\mathbf{x}^{(t)}).$$



Newton Method for Burger's Equation

Let $\alpha_j = -\sigma - \frac{k}{2h}z_j$, $\beta_j = 1 + 2\sigma + \frac{k}{2h}(z_{j+1} - z_{j-1})$, and $\gamma_j = -\sigma + \frac{k}{2h}z_j$. Then J is given by [page 418]

$$J = \begin{bmatrix} 1 & 0 & 0 & 0 & \dots & \dots & 0 \\ \alpha_1 & \beta_1 & \gamma_1 & & & & 0 \\ 0 & \alpha_2 & \beta_2 & \alpha_2 & & & \vdots \\ \vdots & & \ddots & \ddots & \ddots & & \vdots \\ \vdots & & & \ddots & \ddots & \ddots & 0 \\ 0 & \dots & & \alpha_{n-1} & \beta_{n-1} & \gamma_{n-1} \\ 0 & \dots & 0 & 0 & 0 & 1 \end{bmatrix}$$

(See notes.)



Nonlinear PDEs: Fisher's Equation

Another slight modification of the heat equation yields a model for the diffusion of population genetics

Definition

Fisher's equation is given by $u_t = Du_{xx} + g(u)$, where g is a polynomial in u .

Lemma

$x = C$ is an equilibrium to Fisher's equation wherever $g(C) = 0$ and a stable equilibrium when also $g'(C) < 0$.



Fisher's Equation

The Fisher's equation with **Neumann** boundary conditions is given by:

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + g(u)$$

$$u(x, 0) = f(x), \quad x \in [a, b] \quad \text{initial profile}$$

$$u_x(a, t) = l(t), \quad t \in [t_0, T] \quad \text{derivative at left boundary}$$

$$u_x(b, t) = r(t), \quad t \in [t_0, T] \quad \text{derivative at right boundary}$$

Solving with Finite Difference Method

Using backward difference for the the left-hand side, we obtain

$$\frac{u_{i,j} - u_{i,j-1}}{k} + \mathcal{O}(k) = D \left[\frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2} + \mathcal{O}(h^2) \right] + u_{i,j} - u_{i,j}^2$$

which simplifies to

$$\begin{aligned} u_{i,j} &= \sigma[u_{i+1,j} - 2u_{i,j} + u_{i-1,j}] + k[u_{i,j} - u_{i,j}^2] + u_{i,j-1} \\ 0 &= (1 + 2\sigma - k + ku_{i,j})u_{i,j} - \sigma u_{i+1,j} - \sigma u_{i-1,j} + u_{i,j-1}. \end{aligned}$$

To consider the left/right boundaries, we must have

$u_x(0) = u_x(1) = 1$. We use a 2nd-order approximation for the derivatives:

$$\begin{aligned} \frac{\partial}{\partial x} u(0, t_j) &= \frac{-3u_{0,j} + 4u_{1,j} - u_{2,j}}{2k} + \mathcal{O}(k^2) \\ \frac{\partial}{\partial x} u(1, t_j) &= \frac{-3u_{n,j} + 4u_{n-1,j} - u_{n-2,j}}{2k} + \mathcal{O}(k^2). \end{aligned}$$

Newton Method for Fisher's Equation

Let $\beta_j = 1 + 2\sigma - k + 2kz_j$. Then J is given by [page 422]

$$J = \begin{bmatrix} -3 & 4 & -1 & 0 & \dots & \dots & 0 \\ -\sigma & \beta_1 & -\sigma & & & & 0 \\ 0 & -\sigma & \beta_2 & -\sigma & & & \vdots \\ \vdots & & \ddots & \ddots & \ddots & & \vdots \\ \vdots & & & \ddots & \ddots & \ddots & 0 \\ 0 & \dots & & & -\sigma & \beta_{n-1} & -\sigma \\ 0 & \dots & & 0 & -1 & 4 & -3 \end{bmatrix}$$



Solving Fisher's Equation with Backward Finite Differences

Example Fisher's equation.

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + u(1 - u)$$

$$u(x, 0) = \sin(\pi x), \quad x \in [0, 1] \quad \text{initial profile}$$

$$u_x(a, t) = 0, \quad t \in [0, 3] \quad \text{derivative at left boundary}$$

$$u_x(b, t) = 0, \quad t \in [0, 3] \quad \text{derivative at right boundary}$$

Let $D = 1$, $h = k = 0.01$.

(See code.)